



Stability prior to surgery in Congenital Diaphragmatic Hernia: Is it necessary?

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Abstract

Background: Delaying surgery for infants with CDH until they achieve clinical stability is common practice. Stability, however, is inconsistently defined, and many infants fail to reach pre-established criteria. We sought to determine if infants undergoing surgery without meeting pre-established criteria could achieve meaningful survival.

Methods: All infants in the CAPSNet database were analyzed (2005–2010). Patients undergoing operative repair were divided into two groups based on whether they met strict ($\text{FiO}_2 < 0.40$, conventional ventilation, preductal saturation $> 92\%$, no inotropes or vasodilators), or lenient ($\text{FiO}_2 < 0.60$, conventional ventilation, preductal saturation $> 88\%$, no vasodilators) criteria. Univariate analyses were performed comparing characteristics of those who survived after surgery ($N = 273$) with those who did not ($N = 21$).

Results: 294 patients (85%) survived to surgery. Predictors of post-operative survival included prenatal liver position ($p = 0.003$), preoperative oxygen requirements ($p = 0.008$), preoperative inotropes ($p < 0.0001$), and non-conventional ventilation ($p = 0.004$). Infants meeting strict criteria had increased survival (99%; $p < 0.0001$). Infants meeting lenient criteria constituted 70% of survivors. Nearly one-third of survivors met neither strict nor lenient criteria.

Conclusions: Infants with CDH can achieve good survival even when criteria for pre-operative stability are not met. We suggest that all infants should be repaired even if lenient criteria for ventilatory, inotrope, or vasodilator requirements are not achieved.

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Congenital diaphragmatic hernia (CDH) is a rare malformation with an incidence of approximately 1 in 3000 live births [1]. The management of infants with CDH has evolved over the years to the currently widely accepted

practice of delaying surgery until clinical stability has been achieved [2]. However, the definition of “stability” to be achieved prior to surgery is inconsistent and many infants may fail to meet pre-established criteria [3–7]. Whether a subgroup of infants who are not surgical candidates can be identified is of great interest to those caring for this patient population. One of the most common reasons cited for not offering surgery or Extracorporeal Membrane Oxygenation

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(ECMO) is the presence of lethal pulmonary hypoplasia [8–10]. This is poorly defined but generally accepted as being present in infants who fail to meet physiologic criteria [4,11,12].

Many institutions have established protocols for the management of their CDH patients, and implementation of these protocols has resulted in improved outcomes [2,7,13,14]. However, strict adherence to protocols may result in some infants not being offered surgery because of their failure to achieve physiologic targets.

The purpose of this study was to determine if infants undergoing surgery who have not met pre-established guidelines prior to operation could achieve meaningful survival. In effect we are asking: should we have contraindications to surgery based on estimates of pulmonary hypoplasia?

1. Methods

We analyzed all patients with CDH in the CAPSNet database from April 2005 to December 2010. Infants in the database who survived to surgical repair were included in the analysis. Data abstracted included: prenatal information (prenatal diagnosis, liver position, side of defect), perinatal data (outborn or inborn, APGAR scores, birth weight, sex, gestational age, chromosomal anomaly, associated cardiac anomaly), pre-operative status (use of vasopressors, pulmonary vasodilators, oxygen requirements, mode of ventilation, ECMO use, cardiorespiratory status, SNAP-II score) and outcomes (discharge status and destination, oxygen requirement at discharge, length of stay).

We used recommendations for preoperative stability in keeping with those described by Boloker et al. [11] as a guideline to define our criteria for preoperative stability: “Surgical repair was delayed until the preductal to postductal saturation gradient had subsided, echocardiographic examination confirmed minimal evidence of pulmonary hypertension, and ventilator support had minimized.” We established the criteria for infants who would meet either strict or lenient definitions of stability based on a consensus review of the authors as well as recommendations from the CAPSNet Steering Committee. The “strict” group was defined as adherence to the following criteria:

- preductal saturation >92%,
- conventional ventilator settings
- fraction of inspired oxygen <40% ($\text{FiO}_2 < 0.40$),
- not requiring Nitric Oxide or other pulmonary vasodilators
- no vasopressor support.

A “lenient” group was defined as adherence to the following criteria:

- preductal saturation >88%,
- $\text{FiO}_2 < 0.60$,

- conventional ventilator settings,
- not requiring Nitric Oxide or other pulmonary vasodilators.

As CAPSNet and CNN do not collect information on echocardiographic measures of pulmonary hypertension, the use of nitric oxide or pulmonary vasodilators at the time of surgery was identified as a surrogate marker of elevated pulmonary arterial pressures.

In order to be *eligible* for analysis, patients needed to have data available for at least three of the criteria domains outlined above. Infants included in the “strict” or “lenient” criteria groups, met the above requirements for every measured domain, whereas those who did not meet criteria for one or more of the measured domains were classified as not meeting criteria.

1.2. Outcomes and statistical analyses

The primary outcome of interest was mortality prior to discharge. A secondary outcome measured was length of stay (LOS) for survivors. Univariate analysis was used to compare baseline characteristics and pre-operative stability criteria in those infants surviving surgery in each group and was compared to those who did not. LOS was also assessed by univariate analysis. To assess the independent effect of meeting or not meeting the criteria of operative stability, two separate logistic regression analyses were performed looking at the independent effect of meeting either strict or lenient criteria on post-operative survival while controlling for sex, gestational age, SNAP-II score, prenatal liver position, outborn status and the presence or absence of cardiac anomalies (the presence of a chromosomal anomaly was not included as the number of infants was too low for analysis). SNAP-II is an aggregate score which measures the magnitude of physiological derangement of the respiratory, cardiovascular, renal, and central nervous systems, recorded within the first 12 h of NICU admission. SNAP-II has been validated as a predictor of mortality in a number of NICU patient cohorts including CDH [15,16].

As well, linear regression analyses were performed using the same predictive variables to evaluate the independent association of meeting operative stability criteria on length of stay. All analyses were carried out using SAS (version 9.3, 2010 Cary, NC).

2. Results

A total of 348 infants with CDH were reviewed from the CAPSNet database, of whom 294 (84.5%) survived to surgical repair. Fifty patients died prior to surgery, 1 misdiagnosed patient went home without surgery and 2 were transferred to another hospital (1 for palliation). Baseline characteristics for all patients in the database, as well as for the patients surviving to surgery are shown in Table 1. Infants who met inclusion criteria for either the strict

or the lenient group had a significantly higher chance of being outborn (strict 71.6% vs. 57.7% $p=0.02$; lenient 70.5% vs. 44.4% $p<0.0001$), as well as not having the liver in the thoracic cavity (strict 8.8% vs. 28.1% $p=0.03$; lenient 14.5% vs. 35.6% $p=0.01$). Infants meeting strict or lenient criteria also had a significantly lower SNAP-II (Score of Neonatal Acute Physiology). The median SNAP-II for infants meeting strict criteria compared to those not meeting strict criteria was 5 vs. 16 ($p<0.0001$) while the median SNAP-II for infants meeting lenient criteria compared to those not meeting lenient criteria was 10 vs. 16 ($p=0.0003$) (Table 1). Characteristics such as male sex, a right-sided defect or the presence of any chromosomal anomaly were similar for infants meeting or not meeting either strict or lenient criteria.

Of the 294 infants surviving to surgical repair, 291 were eligible for analysis (having data available for three or more criteria available for possible inclusion into the strict group). Of these, 102 met inclusion criteria ($\text{FiO}_2 < 0.4$, conventional ventilation, no vasopressors or pulmonary vasodilators, no ECMO, preductal saturation $> 92\%$), while 189 did not. For the lenient group, 280 patients were eligible for consideration. One hundred and ninety patients met inclusion criteria ($\text{FiO}_2 < 0.6$, conventional ventilation, no pulmonary vasodilators, preductal saturation $> 88\%$), while 90 did not.

For those undergoing surgery, 273 survived post-operatively, while 21 patients died (post-operative mortality of 7.1%). The baseline and pre-operative characteristics of those surviving post-operatively were compared to those who did not using univariate analysis and are presented in Table 2. The presence of the liver in the chest on prenatal ultrasound and SNAP-II score were the only significant baseline characteristics predictive of post-operative death ($p=0.003$ and $p<0.0001$ respectively). Pre-operative criteria that were significant predictors of non-survival post-operatively included a higher median pre-operative FiO_2 ($p=0.008$), the patient being on vasopressor support at time of operation ($p<0.0001$), and the patient requiring non-conventional ventilation, such as high frequency oscillation, jet ventilation or hand-bagging ($p=0.004$).

The survival of CDH infants meeting the strict and lenient criteria was 99% (101/102) and 94.2% (179/191), respectively. When analyzing the 291 patients *eligible* for inclusion in the strict group, there were 270 survivors yet 169 of these survivors (62.6%) did not actually *meet* the strict criteria for surgery. Furthermore, of the 280 patients *eligible* for inclusion in the lenient group, there were 259 survivors of whom 80 (30.9%) also did not *meet* the lenient criteria for surgery. As suggested by the above findings, logistic regression analyses demonstrated that the failure to meet either strict or lenient criteria did not independently predict survival to discharge when controlling variables linked to CDH survival including gestational age, sex, SNAP-II, side of defect, and prenatal liver position.

LOS was significantly shorter for surviving infants meeting strict criteria compared to those who did not on crude analysis (21.5 days vs. 36.5 median days in hospital; $p<0.0001$). LOS was also shorter for those patients meeting lenient criteria compared to those that did not (27 vs. 41.5 median days in hospital; $p<0.0001$). While linear regression analyses failed to demonstrate that meeting lenient operative criteria was independently associated with a prolonged length of stay, the failure to meet strict operative criteria was independently associated with an additional 27 days in hospital ($p=0.04$).

A total of 16 patients who survived to operative repair underwent ECMO at some time in the course of their therapy. No patients meeting strict criteria underwent ECMO at any time during their course in hospital. Eight patients meeting lenient criteria were treated with ECMO, 4 of whom died (one of whom was on ECMO at the time of surgery).

3. Discussion

There is widespread recognition that a subset of CDH infants exists who have lethal pulmonary hypoplasia and for whom surgery may not be warranted [17]. These infants are

Table 1 Baseline characteristics for all infants with CDH with a subset analysis of those surviving to surgery.

Baseline characteristics	All infants in CDH database (n=348)	Infants surviving to surgery (n=294)					
		Infants meeting strict criteria (n=102)	Infants not meeting strict criteria (n=189)	p-value	Infants meeting lenient criteria (n=190)	Infants not meeting lenient criteria (n=90)	p-value
Male (%)	205 (59.2)	56 (54.9)	114 (60.3)	0.38	111 (58.4)	54 (60.0)	0.90
Right-sided Defect (%)	23 (12.8)	1 (2.6)	13 (12.3)	0.11	7 (7.9)	7 (13.2)	0.38
Liver in Chest (%)	42 (27.8)	3 (8.8)	25 (28.1)	0.03	11 (14.5)	16 (35.6)	0.01
Chromosomal Anomaly (%)	4 (1.4)	0 (0)	3 (1.9)	0.55	1 (0.6)	2 (2.7)	0.23
Cardiac Abnormality (%)	33 (9.5)	9 (8.8)	11 (5.8)	0.34	12 (6.3)	6 (6.7)	1.0
Outborn (%)	209 (59.9)	73 (71.6)	109 (57.7)	0.02	134 (70.5)	40 (44.4)	<0.0001
SNAP-II ^a - median (IQR)	16 (5,30)	5 (0,14)	16 (5,29)	<0.0001	10 (0,23)	16 (5,32)	0.0003

Infants with data available for three or more variables used to determine adherence to strict or lenient criteria were included in the analyses. Proportions and medians are reported for non-missing variables only.

^a SNAP-II: Score for Neonatal Acute Physiology [23].

Table 2 Univariate analysis comparing characteristics of the surviving and non-surviving 294 infants undergoing surgery.

Baseline characteristics	Infants surviving after OR (n=273)	Infants not surviving after OR (n=21)	p-value
Male (%)	158 (57.9)	14 (63.6)	0.66
Right-sided defect (%)	11 (8.6)	3 (17.7)	0.21
Liver in Chest on Prenatal US	20 (18.4)	8 (57.1)	0.003
Median Gestational Age (IQR)	38 (37,39)	38 (36, 40)	0.39
Median Snap Score (IQR)	12 (0,23)	29 (23,32)	<0.0001
Presence of chromosomal anomaly (%)	2 (0.9)	1 (5.3)	0.22
Presence of Major Cardiac abnormality (%)	18 (6.6)	2 (9.5)	0.64
Outborn (%)	173 (63.4)	11 (50.0)	0.25
Preoperative Characteristics			
Median Pre-Op Fraction of Inspired Oxygen (IQR)	30 (24, 37)	41 (30, 50)	0.008
Median Pre-Op Preductal Oxygen Saturation (IQR)	97 (95, 99)	98.5 (94.5, 99)	0.68
Patients on inspired Nitric Oxide at the time of OR (%)	31 (11.5)	5 (23.8)	0.16
Patients on other vasodilator at the time of OR (%)	15 (5.6)	2 (9.5)	0.35
Patients on pressors at time of OR (%)	135 (49.5)	19 (90.5)	<0.0001
Patient on HFO, jet ventilator or handbagging at time of OR (%)	38 (15.2)	8 (40.0)	0.01

Proportions and medians given are reported for non-missing variables.

usually identified by their failure to meet pre-established operative criteria [18]. As well as perioperative physiologic measures, baseline characteristics such as the presence of the liver in the chest and higher SNAP-II scores were predictive of increased mortality [16,19,20].

Anecdotally, however, there are infants who do not meet established criteria for clinical stability prior to operative repair, yet they undergo repair and manage to survive. We evaluated the CAPSNet database, which contains prospectively collected data on all infants with CDH across Canada, to determine if the application of strict or lenient operative criteria could identify a subpopulation of infants unlikely to survive surgery.

Strict criteria for preoperative stability identify those infants with the best chance of survival, and our results demonstrate that 99% of these infants survive post-operatively. However, infants meeting the strict criteria represent only one-third of all CDH survivors. If the stability criteria are relaxed to include admission into the lenient group, the survival rate is still excellent at over 90%. Interestingly, the lenient group represents two-thirds of the survivor cohort, thereby leaving approximately one-third of survivors who do not meet any stability criteria yet have survived after operation. These results suggest that the application of any physiologic stability criteria by clinicians is highly variable. The reasons for this are unclear but may be related to physician attitudes and bias. Moreover, these findings underscore reports identifying significant practice pattern variability and its effect on patient outcome in Canada [21]. The reasons for the wide variability in the application of physiologic criteria to determine surgical “readiness” are unclear. Ideally, every infant with CDH should be treated in a similar fashion using the best “evidence” as a guide. The standardization of care in CDH infants has been linked to improved outcomes in this cohort of patients [7,22].

We chose to restrict our analysis to those patients who underwent surgical repair of their CDH, and compared survival outcomes for those who met or did not meet inclusion criteria for the strict or lenient groups. Infants who do not meet either strict or lenient criteria make up a significant proportion of post-operative survivors. However, there were 53 patients in the database who did not undergo repair. As we restricted our analysis to those infants undergoing surgery, we have likely excluded a significant number of patients who would have failed to meet either strict or lenient criteria for surgery and who died without an operation.

Adherence to strict criteria for pre-operative clinical stability in patients with CDH leads to excellent survival outcomes. Relaxation of these criteria to more lenient parameters is also associated with low mortality rates. However, not offering surgery to patients who may never meet these criteria for pre-operative stability may deny surgery to potential survivors. We have not looked at the long-term survival or morbidity of those infants that fail to meet strict or lenient criteria. Analysis of these outcomes may identify criteria that allow us to distinguish a population of infants who do not benefit from surgery. Until that time, we advocate that stability criteria be used to guide timing of surgery but not be used to deny infants with CDH operative repair.

References

- [1] Logan JW, Rice HE, Goldberg RN, et al. Congenital diaphragmatic hernia: a systematic review and summary of best-evidence practice strategies. *J Perinatol* 2007;27:535-49.
- [2] Wung JT, Sahni R, Moffitt ST, et al. Congenital diaphragmatic hernia: survival treated with very delayed surgery, spontaneous respiration, and no chest tube. *J Pediatr Surg* 1995;30:406-9.

- [3] Frenckner B, Ehren H, Granholm T, et al. Improved results in patients who have congenital diaphragmatic hernia using preoperative stabilization, extracorporeal membrane oxygenation, and delayed surgery. *J Pediatr Surg* 1997;32:1185-9.
- [4] Kays DW, Langham Jr MR, Ledbetter DJ, et al. Detrimental effects of standard medical therapy in congenital diaphragmatic hernia. *Ann Surg* 1999;230:340-8.
- [5] Nakayama DK, Motoyama EK, Tagge EM. Effect of preoperative stabilization on respiratory system compliance and outcome in newborn infants with congenital diaphragmatic hernia. *J Pediatr* 1991;118:793-9.
- [6] Al-Hathlol K, Elmahdy H, Nawaz S, et al. Perioperative course of pulmonary hypertension in infants with congenital diaphragmatic hernia: impact on outcome following successful repair. *J Pediatr Surg* 2011;46:625-9.
- [7] Tracy ET, Mears SE, Smith PB, et al. Protocolized approach to the management of congenital diaphragmatic hernia: benefits of reducing variability in care. *J Pediatr Surg* 2010;45:1343-8.
- [8] Keshen TH, Gursoy M, Shew SB, et al. Does extracorporeal membrane oxygenation benefit neonates with congenital diaphragmatic hernia? Application of a predictive equation. *J Pediatr Surg* 1997;32:818-22.
- [9] Heiss KF, Clark RH. Prediction of mortality in neonates with congenital diaphragmatic hernia treated with extracorporeal membrane oxygenation. *Crit Care Med* 1995;23:1915-9.
- [10] Lally KP. Extracorporeal membrane oxygenation in patients with congenital diaphragmatic hernia. *Semin Pediatr Surg* 1996;5:249-55.
- [11] Boloker J, Bateman DA, Wung JT, et al. Congenital diaphragmatic hernia in 120 infants treated consecutively with permissive hypercapnea/spontaneous respiration/elective repair. *J Pediatr Surg* 2002;37:357-66.
- [12] Seetharamaiah R, Younger JG, Bartlett RH, et al. Factors associated with survival in infants with congenital diaphragmatic hernia requiring extracorporeal membrane oxygenation: a report from the Congenital Diaphragmatic Hernia Study Group. *J Pediatr Surg* 2009;44:1315-21.
- [13] Bagolan P, Casaccia G, Crescenzi F, et al. Impact of a current treatment protocol on outcome of high-risk congenital diaphragmatic hernia. *J Pediatr Surg* 2004;39:313-8.
- [14] Finer NN, Tierney A, Etches PC, et al. Congenital diaphragmatic hernia: developing a protocolized approach. *J Pediatr Surg* 1998;33:1331-7.
- [15] Gagliardi L, Cavazza A, Brunelli A, et al. Assessing mortality risk in very low birthweight infants: a comparison of CRIB, CRIB-II, and SNAPPE-II. *Arch Dis Child Fetal Neonatal Ed* 2004;89:F419-22.
- [16] Skarsgard ED, MacNab YC, Qiu Z, et al. SNAP-II predicts mortality among infants with congenital diaphragmatic hernia. *J Perinatol* 2005;25:315-9.
- [17] Gentili A, De Rose R, Iannella E, et al. Is the time necessary to obtain preoperative stabilization a predictive index of outcome in neonatal congenital diaphragmatic hernia? *Int J Pediatr* 2012;40:170.
- [18] Bohn D, Tamura M, Perrin D, et al. Ventilatory predictors of pulmonary hypoplasia in congenital diaphragmatic hernia, confirmed by morphologic assessment. *J Pediatr* 1987;111:423-31.
- [19] Mayer S, Klaritsch P, Petersen S, et al. The correlation between lung volume and liver herniation measurements by fetal MRI in isolated congenital diaphragmatic hernia: a systematic review and meta-analysis of observational studies. *Prenat Diagn* 2011;31:1086-96.
- [20] Mullassery D, Ba'ath ME, Jesudason EC, et al. Value of liver herniation in prediction of outcome in fetal congenital diaphragmatic hernia: a systematic review and meta-analysis. *Ultrasound Obstet Gynecol* 2010;35:609-14.
- [21] Baird R, Eeson G, Safavi A, et al. Institutional practice and outcome variation in the management of congenital diaphragmatic hernia and gastroschisis in Canada: a report from the Canadian Pediatric Surgery Network. *J Pediatr Surg* 2011;46:801-7.
- [22] van den Hout L, Schaible T, Cohen-Overbeek TE, et al. Actual outcome in infants with congenital diaphragmatic hernia: the role of a standardized postnatal treatment protocol. *Fetal Diagn Ther* 2011;29:55-63.
- [23] Richardson DK, Corcoran JD, Escobar GJ, et al. SNAP-II and SNAPPE-II: simplified newborn illness severity and mortality risk scores. *J Pediatr* 2001;138:92-100.